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**TOPIC: STUDENT ACADEMIC PERFORMANCE & LIFESTYLE ANALYTICS**

**TITLE: A BUSINESS INTELLIGENCE APPROACH TO UNDERSTANDING HOW LIFESTYLE HABITS, DIGITAL DISTRACTIONS AND MENTAL WELLBEING AFFECT STUDENT PRODUCTIVITY AND ACADEMIC OUTCOMES**

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**1.0 Problem Background**

This section provides an overview of the issue being addressed including the business context, the importance of the problem and the expected outcomes.

**1.1 Business Context**

University students deal with an increasingly complex array of factors that impact their academic achievement in today’s digitally connected environment. In addition to conventional academic considerations like study time or attendance in class, modern students have to deal with widespread smartphone use, social media, gaming habits, irregular sleep cycles and increasing psychological stress. Higher education institutions, academic advisors and student welfare departments must understand these factors to design effective intervention programmes and support systems.

From a Business Intelligence (BI) perspective, students' academic data represents a rich and useful information asset. Data-driven insights that help identify at-risk students early, optimize academic support resources and match curriculum delivery with evidence-based student productivity patterns are beneficial to universities, government education ministries and edtech companies. This project define the academic performance challenge as a business problem : how can institutions use integrated datasets to produce actionable intelligence that enhances student outcomes on a large scale?

**1.2 Importance of the Problem**

Academic underperformance carries serious economic, social and institutional consequences. Inadequate GPAs put students at risk for lower employment, delayed graduation and scholarship loss. When student turnover rates increase, institutions suffer financial and reputational consequences. Every year, student mental health issues alone cost the world of billions of dollars in decreased productivity.

Crucially, current academic reporting systems frequently analyze these dimensions in isolation. For example, digital usage data is rarely connected with academic outcomes and GPA trends are analyzed independently from mental health surveys. Siloed approaches completely ignore hidden correlations and underlying causes that can be uncovered by a unified BI solution that integrates lifestyle, productivity and mental health datasets. For example, a student may put in enough study hours but still perform poorly because of excessive social media use during study periods, poor sleep or high depression scores, which a pattern would be undetectable without data integration.

**1.3 Expected Impact**

The goal of this project is to provide a BI solution that produces actionable, evidence-based insights for academic stakeholders. The expected impact are outlined below:

Table 1.1 Expected impact

| Stakeholder            | Expected Benefit                                                                                                                                 |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Academic Advisor       | Early detection of students exhibiting lifestyle risk factors linked to low GPA - targeted counseling referrals.                                 |
| Student Welfare Units  | Prioritizing mental health resources for students exhibiting high levels of stress and depression based on data.                                 |
| University Management  | Benchmarking institutional performance by lifestyle categories - evidence base for campus policy decisions                                       |
| Students               | Self-awareness dashboards that display the relationship between daily routines and academic performance - advice for individualized improvement. |
| Education Policymakers | Macro-level understanding of how national academic productivity is impacted by mental health and digital distractions.                           |

## 2.0 Objectives

This project is guided by the following research and development objectives:

### 1. Data Integration

To integrate three student datasets (lifestyle, digital productivity and mental health) into a unified, analytics-ready data model using Alteryx Designer, ensuring data consistency, completeness and referential integrity across all combined records.

### 2. Lifestyle Impact Analysis

To analyze the academic outcomes measured by GPA and final examination grades and quantify the relationship between key lifestyle variables including daily study hours, sleep duration, physical activity frequency and social engagement

### 3. Digital Distraction Assessment

To assess the impact of digital distraction factors such as smartphone screen time, social media usage hours and gaming hours on student focus scores, attendance rates and overall productivity scores.

### 4. Mental Wellbeing Correlation

To examine the correlation between academic performance metrics (GPA, final grades) and student mental health indicators including stress levels, depression scores and anxiety makers. This relationship identifies the threshold value where mental wellbeing significantly impacts outcomes

## 5. Student Segmentation

To segment the student population into meaningful performance clusters using BI analytics techniques so that high-risk, moderate-risk and high-performing student cohorts can receive different intervention measures

## 6. Dashboard Development

To design and develop an interactive BI dashboard using Tableau or Power BI that presents multidimensional insights through at least five different visualization types, supporting data-driven decision-making for academic administrators.

## 7. Decision Support

To provide academic institutions with practical, evidence-based suggestions about curriculum scheduling, digital device policies, the distribution of resources for mental health support and focused student intervention initiatives.

### 3.0 Datasets Description

This project utilises three publicly available, open-source datasets sourced from Kaggle. Each dataset contributes a distinct analytical dimension to the integrated Business Intelligence (BI) solution. Together, the datasets provide over 122,000 student records spanning lifestyle habits, academic outcomes, digital behaviour, and mental health indicators. Table 3.1 below summarises the datasets used in this study, including their sources, data types, record counts, and key variables.

Table 3.1 : Summary of Datasets Used in the Study

| Dataset       | Student Lifestyle Dataset                                                                                                                                                                      | Student Productivity & Digital Distraction Dataset                                                                                                                                                                                       | Student Depression & Lifestyle Dataset                                                                                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source        | Kaggle – Steve1215rogg<br>Link : <a href="https://www.kaggle.com/datasets/steve1215rogg/student-lifestyle-dataset">https://www.kaggle.com/datasets/steve1215rogg/student-lifestyle-dataset</a> | Kaggle – Sehaj1104<br>Link : <a href="https://www.kaggle.com/datasets/sehaj1104/student-productivity-and-digital-distraction-dataset">https://www.kaggle.com/datasets/sehaj1104/student-productivity-and-digital-distraction-dataset</a> | Kaggle – Aldinwhyudii<br>Link : <a href="https://www.kaggle.com/datasets/aldinwhyudii/student-depression-and-lifestyle-100k-data">https://www.kaggle.com/datasets/aldinwhyudii/student-depression-and-lifestyle-100k-data</a> |
| Record Count  | Approximately 2000 student records                                                                                                                                                             | Approximately 20000 student records                                                                                                                                                                                                      | Approximately 100000 student records                                                                                                                                                                                          |
| Data Type     | Structured, tabular CSV data                                                                                                                                                                   | Structured, tabular CSV data                                                                                                                                                                                                             | Structured, tabular CSV data                                                                                                                                                                                                  |
| Key Variables | Study hours per day, sleep duration (hours), social activity frequency,                                                                                                                        | Daily phone screen time (hours), gaming hours per day, social media usage                                                                                                                                                                | CGPA (Cumulative Grade Point Average), perceived stress level, daily sleep                                                                                                                                                    |

|  |                                                                                   |                                                                                                                                |                                                                                                                                     |
|--|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|  | physical activity level, stress level (categorical), GPA (numeric, 0.0–4.0 scale) | (hours), class attendance percentage, focus score (0–100), final examination grade (A–F), overall productivity score (numeric) | hours, daily social media usage (hours), physical activity frequency, study habits score, depression indicator (binary/categorical) |
|--|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

## 4.0 Proposed BI Approach

This section outlines the general Business Intelligence (BI) methodology employed in this research to examine the connection between academic achievement, mental health and student lifestyle. The method combines data processing, data engineering and visualization approaches to turn raw data into insightful knowledge. The project seeks to facilitate efficient analysis and clear communication of results through the use of suitable tools and organized procedures.

### 4.1 Role of Alteryx and Visualization tools

Throughout this project, Alteryx Designer will be the primary platform for data engineering and transformation because of its drag-and-drop workflow environment, which is ideal for combining various data sources. This is especially helpful for managing the three datasets of different sizes to enable effective data preparation without complicated coding. Firstly, all three CSV files will be ingested into the process using the Input Data tools. The Data Cleansing tool will then be used to standardize field formats, eliminate null values, correct data types and trim unnecessary whitespace to guarantee uniformity across all datasets.

Next, the Select tool will be used to filter, rename and rearrange columns so that, prior to integration, all datasets follow the same schema. The Join and Union tools will be utilized to merge the data, unioning records from various sources to facilitate aggregate analysis and connecting datasets based on shared lifestyle factors. The datasets' analytical usefulness will also be increased by using the Formula tool to produce derived variables like a composite Lifestyle Risk Score, categorized GPA bands (Low, Medium and High) and digital distraction intensity levels.

Lastly, the Summarize tool will be used to create group-level aggregations based on performance bands, lifestyle categories and student segments to allow insightful analysis. The end-to-end data engineering workflow will be completed when the processed data is exported in CSV or Excel formats using the Output Data tool to prepare it for further visualization and analysis in programs like Tableau or Power BI. These visualization tools link to provided datasets and transform them into interactive graphs, charts and dashboards.

### 4.2 Data Preparation, Data Integration and Data Transformation

This stage focuses on preparing the collected datasets to ensure they are consistent, clean and ready for analysis. The three primary procedures involved are data preparation, data integration and data

integration. These procedures are crucial for enhancing data quality, integrating various data sources and producing significant variables that facilitate efficient analysis and visualization.

#### **4.2.1 Data Preparation (Cleaning)**

Data preparation involves standardizing and cleaning the datasets to ensure consistency and accuracy. This entails fixing inconsistencies, handling missing values and eliminating duplicate and unnecessary records. The dataset gets more dependable and prepared for additional integration and analysis by carrying out these procedures. The processes listed below are used to clean and standardize the dataset:

- Standardize column names using a standard naming scheme.
- Convert stress levels and physical activity fields from categorical to ordinal numerical values for consistent analysis.
- Handle missing data by using mean or mode imputation for continuous or categorical data respectively.
- Remove duplicate entries and statistical outliers.

#### **4.2.2 Data Integration**

Data integration creates a single unified dataset for analysis by combining several datasets. This procedure makes sure that common variables are used to correctly link related data from various sources. These are the following steps conducted in data integration:

- Perform join on shared variables between Datasets 1 and 3 to create a composite Lifestyle-Mental Health profile.
- Use a GPA/grade range lookup to add Dataset 2 productivity metrics to the integrated profile.
- Create a Master Analytics with consistent data that includes all relevant variables from each of the three sources.

#### **4.2.3 Data Transformation**

Data transformation involves introducing new variables and making changes to existing data in order to facilitate analysis. This includes preparing summary values, classifying data and creating relevant indicators. These transformations make complex data easier to understand and more appropriate for interpretation and visualization. The following steps are used to transform the data into formats that are helpful for analysis:

- Derive a composite Lifestyle Risk Index using weighted combinations of sleep duration, stress level, physical activity and social media overuse.
- Classify students into three performance groups: High Performance ( $GPA \geq 3.5$ ), Moderate Performers ( $GPA = 2.5 - 3.49$ ) and At-Risk Students ( $GPA < 2.5$ )
- Generate aggregated summary tables with average metrics by lifestyle category and performance tier for dashboard consumption.

### **4.3 Analysis Planned and Visualization Forms**

This section outlines the primary analyses to be carried out on the dataset as well as the appropriate visualization formats that will be used to present the findings. Table 5.1 shows the analysis intended to

investigate how student lifestyle characteristics, mental health and academic achievement are related and the chosen visualizations help in the clear communication of trends, patterns and comparison.

Table 4.1 Analysis type and visualization forms

| Type of Analysis                           | Description                                                                                                                                 | Visualization forms |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Correlation Analysis                       | Analyze how study habits affect academic success by examining the correlation between study hours and GPA/CGPA.                             | Scatter Plot        |
| Comparative Group Analysis                 | Examine average GPA, focus scores and depression rates among student groups according to physical activity and stress levels.               | Grouped Bar Chart   |
| Mental Health-Academic Performance Mapping | Analyze the impact of students' mental health conditions (such as depression indicators) on their CGPA and GPA.                             | Box Plot            |
| Trend Analysis                             | Analyze how productivity scores, phone usage and study habits vary over time or across usage levels.                                        | Line Chart          |
| Risk Score Distribution Analysis           | Determine the concentration of low-, medium- and high-risk groups among students by analyzing the distribution of the Lifestyle Risk Index. | Histogram           |